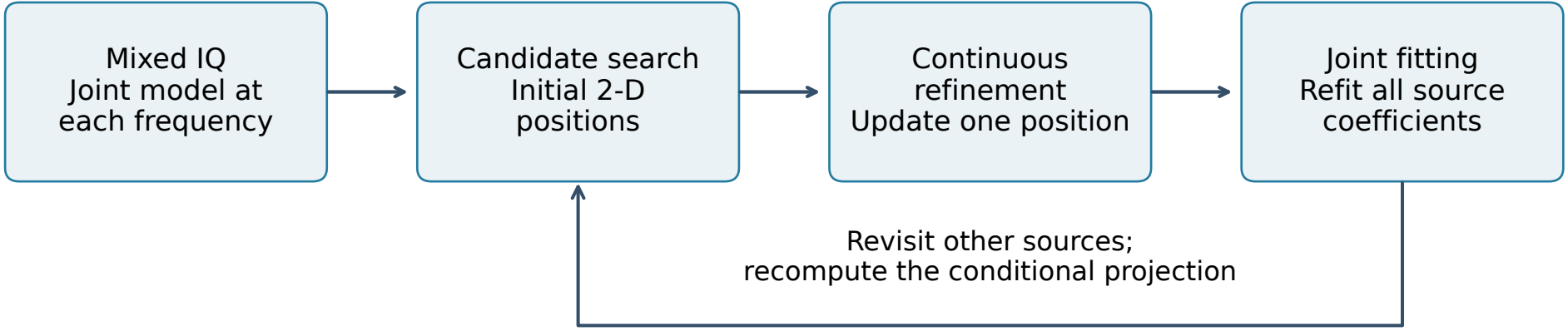Common objective: minimize the joint residual of the original mixture